

Amount [SimTalk] - AGVPool

Sets the amount of AGVs which the *AGVPool* designated by <Path> creates.

Remarks

Plant Simulation creates the specified **Amount** of **AGVs** during the **init phase** of the simulation run.

Type

Attribute

Syntax

```
<Path>.Amount:integer
```

Assignment Value

You can assign a value of data type *integer*.

Example

```
MyAGVPool.Amount := 2
```

See also

[Amount \[text box\] - AGVPool](#)

[AGV \[SimTalk\]](#)

Marker

Marker

Use the object *Marker* for setting waypoints in your simulation model along which the *Automated Guided Vehicle* drives from the *AGVPool* to its *destination*.

Image



Description

With the setting **Default Curve Radius/DefaultCurveRadius** you can set the size of the roundings at the *Markers*.

A **Default Curve Radius** of 0 makes the AGV rotate on the spot instead of driving a curve.

Note

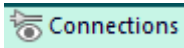
When you insert *Markers* into your model, make sure that the *Markers* are **aligned**. This prevents *Plant Simulation* from computing unnecessary roundings between the *Markers*. Misaligned *Markers* might occur if you inserted *Markers* into your model while **Show Grid** was off.

Note

If a standing *Transporter* cannot drive to the first *Marker* using the specified **Default Curve Radius**, the *Transporter* rotates on the spot, if possible, and then drives in a straight line toward the *Marker*.

The *Marker* can be **directional or omnidirectional**. You can switch between these two states with the check box **Use Rotation of Marker**.

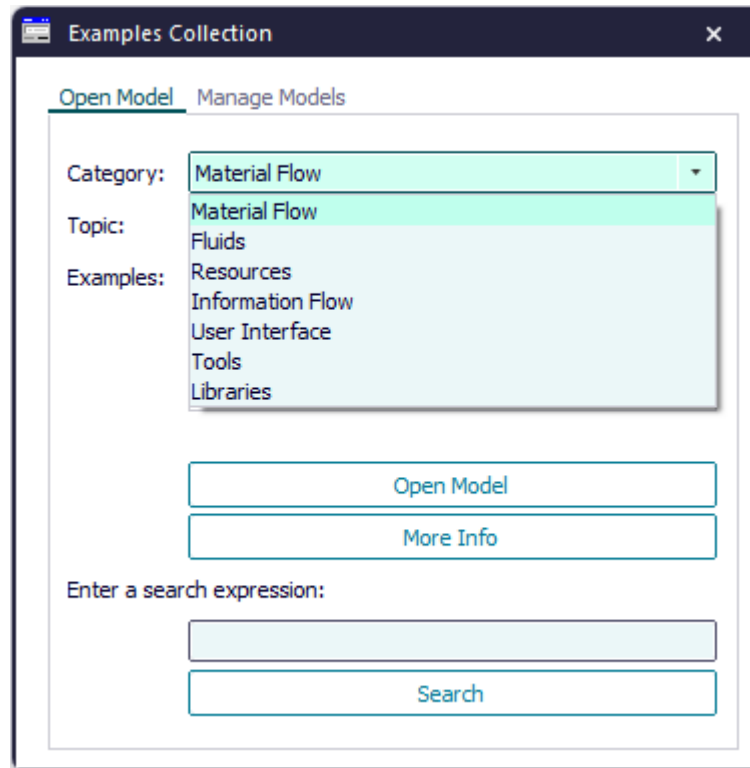
Note

To show or hide a *Marker*, click  **Show Connections**.

Add the Object to the Simulation Model

To add the object *Marker* to your simulation model, click  **Manage Class Library > Basic Objects > Resources > Marker** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Directional and Omidirectional Markers](#)

[Dialog Box of the Marker](#)

[Model an Automated Guided Vehicle System \(AGVS\)](#)

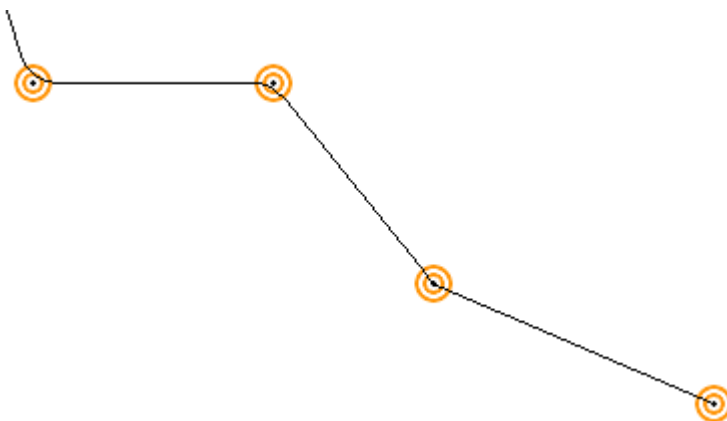
Directional and Omidirectional Markers

The *Markers* in your simulation model can be **directional** or **omnidirectional**.

Remarks

You can switch between these two states with the check box **Use Rotation of Marker**.

- The AGV does not drive through an inserted **omnidirectional** *Marker* but turns in front of the inserted *Markers* toward the next *Marker*. The icon of an *omnidirectional marker* looks like this .



- The AGV can drive through **directional** *Markers* in both directions. For selecting the direction, *Plant Simulation* uses position of the previous *Marker* in the sequence of *Markers*.

Plant Simulation computes the route by means of a *Marker* with the 90-degree angle as follows:

- If, for example, the previous *Marker* is located above the *Marker* with the 90-degree angle, the AGV drives through the *Marker* with the 90-degree angle from top to bottom.
- If the previous *Marker* is located below the *Marker* with the 90-degree angle, the AGV drives through the *Marker* with the 90-degree angle from bottom to top.
- If the previous *Marker* is located exactly at the height of the *Marker* with the 90-degree angle, *Plant Simulation* evaluates the current direction of the AGV.

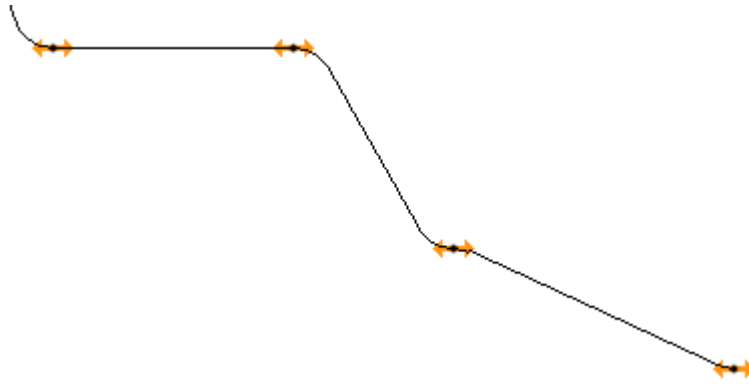
If the front of the AGV points up, the AGV drives through the *Marker* with the 90-degree angle from top to bottom.

If the front of the AGV points down, the AGV drives through the *Marker* with the 90-degree angle from bottom to top.

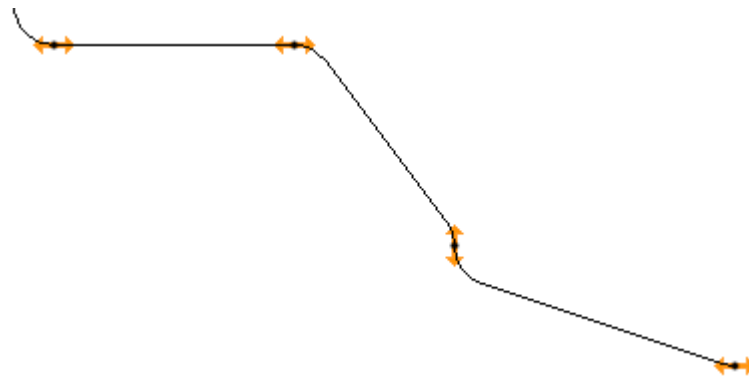
If the front of the AGV points exactly to the center of the *Marker* with the 90-degree angle, the AGV drives through the *Marker* with the 90-degree angle from top to bottom.

For *Markers* with other angles *Plant Simulation* proceeds accordingly.

For a **directional** *Marker* with the default angle of 0° the AGV drives through the *markers* and then turns on the *marker* around the angle you set in the direction of the next *marker*. The icon of a *directional marker* looks like this .



For a **directional Marker** with another than the default angle the AGV drives toward the *Marker* at the angle you entered, crosses it, and leaves it at the designated angle. In the example below we used an angle of 90° . The icon of a *directional marker* with an angle of 90° looks like this .



See also

[Marker](#)

Dialog Box of the Marker

Dialog Box of the Marker

Double-click the icon of the *Marker* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Use Rotation of Marker [check box]

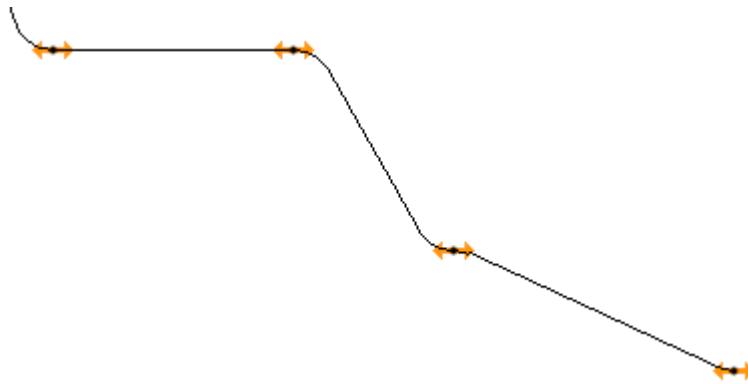
To make the AGV use the rotation of the *Marker* on its way to its destination, select this check box.

☐ Use Rotation of Marker ☒

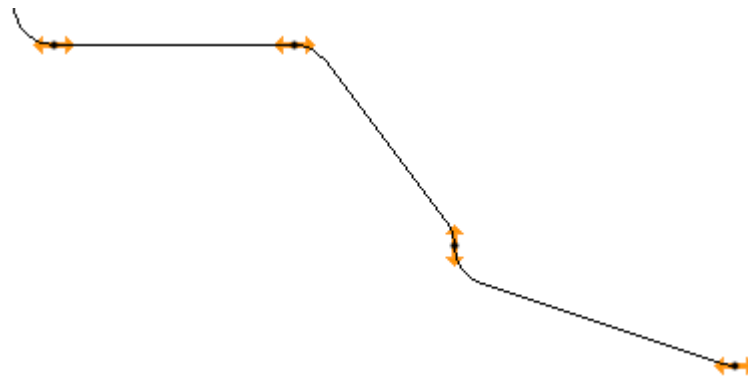
Remarks

Select the check box, to make the AGV drive across the *Marker* and rotate on the *Marker* with the defined angle in the direction of the next *Marker*. The graphic of a *directional marker*, which you inserted into the *Frame*, looks like this .

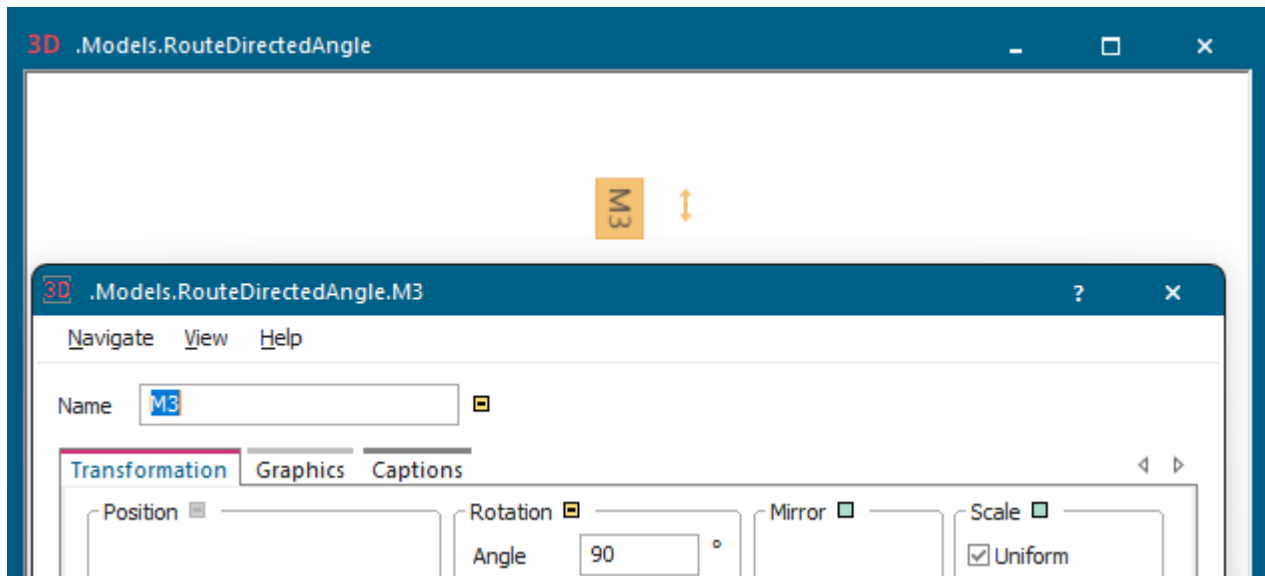
For the default angle of 0° it looks like this:



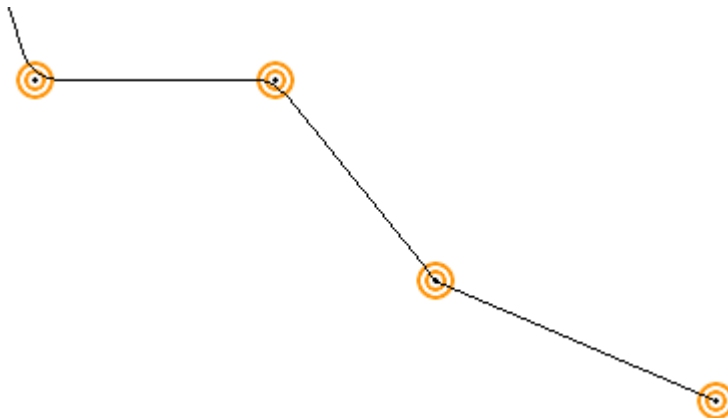
If you type in a rotation angle, for example 90°, it looks like this:



You can set the rotation angle in the dialog **Edit 3D Properties** of the *Marker* on the tab **Transformation** under **Settings for the Rotation**.



Clear the check box ☐ Use rotation of marker ☒ to make the AGV not drive across the inserted *Markers* but rotate in front of the *Marker* toward the next *Marker*. The graphic of an *omnidirectional marker*, which you inserted into the *Frame*, looks like this .



SimTalk

UseRotationOfMarker [SimTalk]

See also

[Model an Automated Guided Vehicle System \(AGVS\) > Cover a Route Along Omnidirectional Markers](#)

[Model an Automated Guided Vehicle System \(AGVS\) > Cover a Route Along Directional Markers](#)

Use Rotation of Marker [check box]

Video on YouTube

https://youtu.be/PYXYutlaGOs?si=6CKYE1f_nLtIVThg&t=924

Arrival Control [Marker]

Modifies the built-in behavior of the object. The object calls the **Arrival Control** as soon as the *Transporter* driving freely within the area arrives at the *Marker*. In the **Arrival Control** you can program actions that you want to take when an AGV arrives at the *Marker* or is closest to the *Marker*. You might provide your fleet management software with feedback about the position of the AGV.



Remarks

For an **omnidirectional** *Marker Plant Simulation* executes the *control* once the **Transporter** driving freely within the area arrives at the half-way point of the curvature/fillet at the *Marker*.

If you did not enter a *Method*, the value of the control is VOID.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



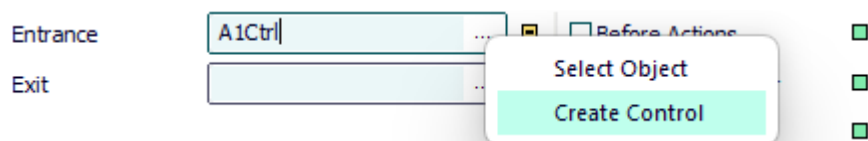
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

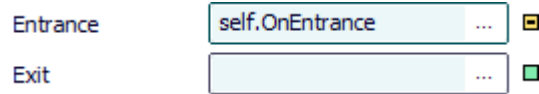
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

[ArrivalCtrl \[SimTalk\]](#)

See also

[Select Object \[for controls\]](#)

Video on YouTube

<https://youtu.be/PYXYutlaGOs?si=FNgErAj2FbcGgyhv&t=409>

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The **View Menu** provides commands to access its functions.

[Refresh \[on View menu\]](#)

[Show Attributes and Methods \[on View menu\]](#)

SimTalk

[updateDialog \[SimTalk\]](#)

See also

[View Menu \[general description\]](#)

Tools Menu

The **Tools Menu** provides these menu commands:

Edit Controls

Edit Observers

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Marker

Methods of the Marker

The *Marker* provides the [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			((Contents:table)) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			((Index:integer:=1)) -> object
MUPart	VOID			((Index:integer:=1)) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).

An example of the **Syntax** line of the individual methods might look like this:

```
<Path>.openDialog([CallOpenControl:boolean:=false]) → boolean
```